Curriculum Vitae

Xiaoshan Li
School of Mathematics and Statistics, Wuhan University,
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Date of Birth April 18, 1985.

Research Interest

Several Complex Variables, CR Geometry and Complex Geometry.

Education

Sep 2008 - Dec 2013	Ph.D. in Mathematics, Wuhan University, China. Advisor: Professor Xiaojun Huang.
Sep 2011 - Sep 2013	Visiting student, Rutgers University, USA.
Sep 2004 - June 2008	B.S. in Mathematics, Wuhan University, China.

Regular position

Dec 2025–Present	Full Professor, Wuhan University, Wuhan, Hubei, China.
Dec 2018–Nov 2025	Associate Professor, Wuhan University, Wuhan, Hubei, China.
May 2016–Nov 2018	Lecture, Wuhan University, Wuhan, Hubei, China.
Oct 2014–Apr 2016	Postdoctoral Fellow, Institute of Mathematics, Academia Sinica, Taiwan
Jan 2014–May 2016	Postdoctoral Fellow, Wuhan University

Visiting Professorship at the Following Institutes

Rutgers University, USA, Aug 2019-July 2020.

Institute of Mathematics, Academia Sinica, Taiwan, Jan 3-Jan31, 2019

University of Cologne, Germany, July 10-Aug 4, 2017.

Institute for Mathematical Sciences, National University of Singapore, Apr 29-May 27, 2017

Institute of Mathematics, Academia Sinica, Taiwan, Jan 9-Jan 22, 2017

University of Cologne, Germany, Dec, 2015.

Awards

ICCM 45-Minutes-Invited Speaker, 2022.

Chutian Scholar of Hubei Province, 2017

Excellent doctoral dissertation of Hubei Province, 2015

Grants

Natural Science Foundation of China-No.12361131577, PI, Jan 2024-Dec 2026; 1, 500, 000.

Natural Science Foundation of China-No. 12271411, PI, Jan 2023-Dec 2026; 450, 000.

Natural Science Foundation of China-No. 11871380, PI, Jan 2019-Dec 2022; 530, 000.

Natural Science Foundation of China-No. 11501422, PI, Jan 2016-Dec 2018; 180, 000.

Postdoctoral Science Foundation of China (First Class)-No. 2015M570660, PI, Aug 2015-May 2016; 80, 000.

Central university research Fund-No. 2042015kf0049, PI, Jan 2015-Dec 2016; 50, 000.

Publications

1. *On the rigidity of proper holomorphic self-mappings of Greene-Krantz-Hartogs domains*, (joint with E. Bi and G. Su) **Sci. China Math.** 69 (2026), no. 6, 1453–1464.

2. *G -invariant Bergman kernel and geometric quantization on complex manifolds with boundary*, (joint with C.-Y. Hsiao, R. Huang and G. Shao), **Math. Ann.** 390 (2024), no. 4, 4889–4930.

3. *Extension of multivalued holomorphic functions on a Stein space*, (joint with X. Huang), **Math. Ann.** 388 (2024), no. 3, 2561–2572.

4. *$\overline{\partial}$-equation on a lunar domain with mixed boundary conditions,* (joint with Xiaojun Huang), **Trans. Amer. Math. Soc.** 368 (2016), no. 10, 6915–6937.

5. *S^1-equivariant Index theorems and Morse inequalities on complex manifolds with boundary*, (Joint with Chin-Yu Hsiao, Rung-Tzung Huang and Guokuan Shao), **J. Funct. Anal.** 279 (2020), no. 3, 108558, 51 pp.

6. *Nonembeddability into a fixed sphere for a family of compact real algebraic hypersurfaces*, (joint with Xiaojun Huang, Ming Xiao), **Int. Math. Res. Not. IMRN** 2015, no. 16, 7382–7393.

7. *Szegő kernels and equivariant embedding theorems for CR manifolds*, (Joint with C.-Y. Hsiao and H. Herrmann), **Math. Res. Lett.** 29 (2022), no. 1, 193–246.

8. *Bergman-Einstein metric on a Stein space with a strongly pseudoconvex boundary* (Joint with X. Huang), **Comm. Anal. Geom.** 31 (2023), no. 7, 1669–1692.

9. *On the stability of equivariant embedding of compact CR manifolds with circle action*, (Joint with Chin-Yu Hsiao and George Marinescu), **Math. Z.** 289 (2018), no. 1-2, 201–222.

10. *Morse inequalities for Fourier components of Kohn-Rossi cohomology on CR manifolds with S^1-action*, (Joint with Chin-Yu Hsiao), **Math. Z.** 284 (2016), no. 1-2, 441-468.

11. *Rado's Theorem for CR Functions on Hypersurfaces*, (joint with S. Berhanu), **J. Geom. Anal.** 34 (2024), no. 10, Paper No. 315.

12. *On a holomorphic family of stein manifolds with strongly pseudoconvex boundaries*, (Joint with Guicong Su), **J. Geom. Anal.** 30 (2020), no. 1, 632–645.

13. *On the Bergman kernel function and weak Morse inequalities*, (joint with G. Shao and H. Wang), **Anal. Math. Phys.** 13 (2023), no. 5, Paper No. 68. 32 (53).

14. *Equivariant Kodaira embedding theorem for CR manifolds with circle action,* (Joint with C.-Y. Hsiao and G. Marinescu), **Michigan Math. J.**, 70 (2021), no. 1, 55–113.

15. *Torus equivariant Szegő kernel asymptotics on strongly pseudoconvex CR manifolds*, (Joint with Hendrik Herrmann and Chin-Yu Hsiao), **Acta Math. Vietnam.** 45 (2020), no. 1, 113–135.

16. *Morse inequalities and embeddings for CR manifolds with circle action*, (joint with Hendrik) **Bull. Inst. Math. Acad. Sin. (N.S.)** 15 (2020), no. 2, 93–122.

17. *Szegő kernel asymptotics and Morse inequalities on CR manifolds with S^1 action*, (Joint with Chin-Yu Hsiao), **Asian Journal of Mathematics**, 22 (3), 413-450, 2018.

18. *Szegő kernel asymptotic expansion on CR manifolds with S^1-action,* (Joint with Hendrik Herrmann and Chin-Yu Hsiao), **Internat. J. Math.** 29 (2018), no. 9, 1850061, 35 pp.

19. *An explicit formula of Szegő kernel on Heisenberg group*, (Joint with Hendrik Herrmann and Chin-Yu Hsiao), **Acta Mathematica Sinica**, English Series, 34 (8), 1225–1247, 2018.

20. *Szegő kernel expansion and embedding of Sasakian manifolds*, (Joint with Hendrik Herrmann and Chin-Yu Hsiao), **Ann. Global Anal. Geom.** 52 (2017), no. 3, 313-340.

21. *Closed range property for $\overline{\partial}$ on Poincaré disk*, **Bull. Inst. Math. Acad. Sin. (N.S.)** 11 (2016), no. 3, 571–578.

22. *The $\overline{\partial}$-equation on an annulus with mixed boundary conditions*, (joint with Mei-Chi Shaw), **Bull. Inst. Math. Acad. Sin. (N.S.)** 8 (2013), no. 3, 399–411.

23. *A simple proof of a theorem of Calabi*, (joint with Xiaojun Huang), **Illinois J. Math.** 56 (2012), no. 1, 95–99 (2013).

Preprints

1. *A note on csc Bergman metric*, (with Xiaojun Huang), arXiv:2606.01319, 2026, submitted

2. ***Localization of Bergman Kernels and the Cheng-Yau Conjecture on Real Analytic Pseudoconvex Domains*, (with Chin-Yu Hsiao and Xiaojun Huang), arXiv:2604.05542, 2026, submitted**

3. *On the Bergman metric of a pseudoconvex domain with a strongly pseudoconvex polyhedral boundary point*, (with Xiaojun Huang and Scott James), arXiv:2512.08275, 2025, submitted

4. *Semi-classical Bergman kernel asymptotics on complex manifolds with pseudo-concave boundary*, (with Chin-Yu Hsiao and George Marinescu), arXiv: 220812412v1, 2022, submitted

Invited Talks

1. *XII Workshop on Geometric Analysis of PDEs and Several Complex Variables*, August 04 to August 08, 2025, in Serra Negra, Sao Paulo, Brazil.

2. *Annual conference of Several Complex Variables*, Zhuhai, 2025.

3. *SCV, CR geometry and Dynamics*, RIMS, Kyoto University, 25–29 November 2024

4. *Shenzhen International Workshop on Several Complex Variables and CR geometry*, Shenzhen, Sep 2–6, 2024.

5. *Special session UMI-AMS Joint Meeting "Geometric Analysis of SCV and PDEs"*, Università degli Studi di Palermo, Palermo Italy, July 25-26, 2024.

6. *Workshop on Several Complex Variables, Complex Geometry and related PDEs*, University of Maryland, USA, May 13-17, 2024.

7. *Annual conference of Several Complex Variables*, Shanghai, 2022.

8. *ICCM 45 minutes invited*, Nanjing, 2022

9. *Seminare on Algebraic Geometry and Complex Analysis,* University of Wuppertal, Jan 21, 2021, Germany

10. *Seminar semiklassische Analysis und Darstellungstheorie,* University of Koln, Dec 1, 2020, Germany

11. *Annual conference of Chinese Mathematican*, 16-19 Oct, 2020, Hebei

12. *The 12th International Conference on Clifford Algebras and Their Applications in Mathematical Physics*, Aug 3-7, 2020, USTC.

13. *AMS Western Sectional Meeting*, Nov 9-10 2019, University of California, Riverside, USA

14. *Seminare on Complex Analysis,* Nov 5, 2019, UCSD, USA.

15. *Taipei Conference on Geometric Invariance and Partial Differential Equations*, Taipei, Jan 17-20, 2018 .

16. *AS-NCTS seminare on several complex variables*, Institute of Mathematics, Academia Sinica, Taipei, Oct 4, 2018.

17. *Annual conference of Several Complex Variables*, Zhejiang, 21-23 August, 2018.

18. *Special session on complex geometry and several complex variables at* **AMS-CMS meeting**, Fudan University, Shanghai, June 11-14 2018.

19. *Symmetries in symplectic, contact and Poisson geometry*, Jiangshu, 8-10 June, 2018.

20. *Taipei Conference on Geometric Invariance and Partial Differential Equations*, Institute of Mathematics, Academia Sinica, Taiwan, Jan 17-20, 2018.

21. *Siyuan Workshop on Geometric Analysis*, Shanghai, 1-3 Dec, 2017.

22. *Young Mathematicians Workshop on Several Complex Variables and Complex Geometry* 2017, Capital Normal University, Dec 9-11 2017.

23. *Several complex variables and complex geometry seminar for young scholars*, Shanghai, 1-3 Dec, 2017.

24. *Taipei Conference on Geometry and Several Complex Variables*, Academia Sinica, Taipei, July 3-7, 2017.

25. *Annual conference of Several Complex Variables, Hebei, August*, 2016, China.

26. *Sinica-NCTS Seminar on Geometry and Several Complex Variables*, Academia Sinica, Taipei, Apr 30, 2016.

27. *Complex Analysis seminar*, University of Cologne, Dec 15, Germany.

28. *Young mathematician workshops on several complex variables*, Tongji University, Aug, 2015.

29. *Seminar on Several Complex Variables, Academia Sinica*, July 2015, Taipei.

30. *Analytic aspects of* $\overline{\partial}$, February 2015, Nagoya University, Japan.

31. *Seminar on several complex variables*, Academia Sinica, Taipei, February 2015.

32. *Seminar on Geometry and Several Complex Variables*, Academia Sinica, Taipei, Oct 2014,